Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau

Phase 6 – Project Documentation

Part A: Final Report

1. Project Summary

Cosmetic Insights is a Tableau-based analytics solution built to help cosmetics businesses and analysts understand market trends, pricing patterns, product distribution, and skin-type coverage. It uses a cosmetics dataset containing brand, product category, price, rank, and skin-type suitability (Dry, Oily, Combination, Normal, Sensitive) to surface actionable insights through an interactive dashboard and a scenario-driven story, embedded in a Flask web application.

2. Objective

To enable stakeholders — brand managers, analysts, and regulatory bodies — to identify high-performing products, monitor consumer preference shifts, respond quickly to product concerns, and anticipate future trends using real-time, data-driven visual analytics.

3. Scope

- Data collection, cleaning, and preparation of the cosmetics dataset.
- Development of 6 unique Tableau visualizations covering pricing, brand, and skin-type analysis.
- A responsive interactive dashboard with Brand, Category, Skin Type, and Price filters.
- A 3-scene story mapped to the three business scenarios.
- Web integration of the dashboard and story via Flask.

4. Key Outcomes

- Real-time visibility into consumer preference trends (Scenario 1).
- Faster identification and management of product concerns (Scenario 2).
- Predictive insight into emerging trends to guide product innovation (Scenario 3).

5. Tools & Technologies Used

Tableau Desktop/Public, Excel/CSV, Flask (Python), HTML/CSS.

6. Conclusion

The Cosmetic Insights project demonstrates how a well-structured Tableau dashboard, backed by clean data and clear business scenarios, can turn raw cosmetics data into a decision-making tool for real-world stakeholders.

Part B: Functional Specification Document (FSD)

1. Purpose

This section defines the functional behavior of the Cosmetic Insights solution — what the system does, for whom, and how each component behaves.

2. Stakeholders

Role	Interest
Brand Marketing Manager	Monitor trends, plan campaigns and product adjustments
Regulatory Body	Track product safety concerns and consumer trust impact
Data Analyst	Explore pricing, brand, and skin-type patterns

3. Functional Modules

Module	Functionality
Data Connection	Connects Tableau to the cosmetics dataset (Excel/CSV)
Data Preparation	Cleans data, pivots skin-type field, creates calculated fields
Visualization Module	Generates 6 chart types covering price, brand, and skin-type views
Dashboard Module	Combines visuals with interactive filters in a responsive layout
Story Module	Presents a 3-scene narrative aligned to the business scenarios
Web Integration Module	Embeds dashboard/story into a Flask web app for stakeholder access

4. Assumptions & Constraints

- The cosmetics dataset is reasonably clean and structured before ingestion into Tableau.
- Tableau Public/Desktop and a stable internet connection are available for publishing.
- The Flask app is used only to display the embedded dashboard/story, not to process data itself.

This completes Phase 6 (Project Documentation), covering the Final Report and the Functional Specification Document (FSD).